

# ZHENG LIU

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Available: Fall 2026 (20–40 hrs/wk, remote) | Graduating 2028 | Green Card Pending

## SUMMARY

Ph.D. student in Computer Science (4.0/4.0) with deep expertise in **generative AI for healthcare**. Built **multi-agent LLM systems for healthcare claims adjudication (F1: 24% → 83%)**, **RAG pipelines with systematic hallucination evaluation (nDCG +63.9%)**, and a **Medicare payment policy compiler processing 2.3M NCCI edits**. Published at ISMB, EMNLP (x2), SIGIR, EAAI Q1 — including healthcare NLP and biomedical AI. **6+ years Python**, hands-on AWS. **Kaggle Expert (top 0.7%)** including **medical imaging**.

## EDUCATION

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| <b>Northeastern University</b><br><i>Ph.D. Computer Science (Full Funding)</i> — GPA: <b>4.0/4.0</b><br>Focus: AI Systems, Biomedical Knowledge Graphs   Advisor: Prof. Benjamin Gyori                                                                     | <i>Boston, MA</i><br><i>Sep 2024 – Present</i>  |
| <b>University College London (UCL)</b><br><i>MSc Knowledge, Information and Data Science</i> — <i>First Class – Distinction</i> — GPA: <b>4.0/4.0</b><br>ML (Distinction), NLP (Distinction), Statistics (High Distinction), Database Theory (Distinction) | <i>London, UK</i><br><i>Sep 2022 – Sep 2023</i> |
| <b>Tongji University (985) &amp; University of Bologna</b> — <i>Full-Tuition Scholarship</i><br>Data Structures & Algorithms, Probability, Linear Algebra, Optimization, Differential Equations                                                            |                                                 |

## TECHNICAL SKILLS

- Generative AI & LLM:** multi-agent architectures, RAG pipelines, LLM fine-tuning (LoRA), structured outputs, prompt engineering, Hugging Face Transformers, vector embeddings (FAISS), hallucination evaluation, chain-of-thought reasoning
- Healthcare Informatics:** CPT/HCPCS & ICD-10 coding, NCCI PTP edits, claims adjudication logic, medical necessity assessment, biomedical knowledge graphs (INDRA), drug target discovery, clinical NLP
- Cloud & Data Engineering:** AWS (Lambda, S3, SQS, DynamoDB, CDK), Docker, CI/CD (GitHub Actions), PostgreSQL, TimescaleDB, data pipeline design, REST APIs (FastAPI)
- Programming & ML:** Python 6+ yrs (Pandas, NumPy, scikit-learn, PyTorch, SciPy), SQL, systematic ablation & factorial experimental design, statistical analysis

## HEALTHCARE AI & GENERATIVE AI RESEARCH

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| <b>PolicyForge — Healthcare Payment Policy Compiler</b>   <i>Independent Project</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>2026 – Present</i>     |
| <ul style="list-style-type: none"><li>Designed an end-to-end system compiling <b>2.3M NCCI PTP edit pairs</b> and <b>4,484 LCD coverage articles</b> from CMS public data into auditable, executable claims rules with <b>character-level citation</b> linking each rule to its source document.</li><li>Built five-stage LLM pipeline with <b>OpenAI Structured Outputs</b> (strict JSON schema) plus a deterministic hallucination gate verifying verbatim substring matches; five-level ablation study: <b>hallucination 66.7% → 0%</b>, <b>HCPC F1 = 0.667</b>. Quarterly NCCI version diff <b>matched official CMS change file exactly</b> across 2,199 additions.</li><li>Impact analysis: PLA molecular/genomic codes = 24% of added edits but drive <b>47% of estimated denial exposure (\$591K quarterly)</b>, directly informing reviewer prioritization. Demonstrates <b>healthcare payment &amp; operations AI</b>.</li></ul> |                           |
| <b>Multi-Agent Clinical Decision Support System</b>   <i>Northeastern University</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>2025 – Present</i>     |
| <ul style="list-style-type: none"><li>Designed a <b>3-agent chain-of-thought pipeline for healthcare claims adjudication</b>: Clinical Analyzer (medical necessity from patient records), Guideline Checker (CPT/ICD-10 cross-reference against structured knowledge base), and Decision Synthesizer (APPROVE/DENY/REFER with confidence scoring and risk flags).</li><li>Implemented <b>multi-model LLM orchestration (GPT-4o / GPT-4o-mini)</b> with agent-to-agent context passing, structured output parsing, and Streamlit audit dashboard. Validated across 3 clinical scenarios (knee arthroplasty, lumbar MRI, cardiac catheterization) with <b>guideline-compliant decisions matching expert adjudication</b>.</li></ul>                                                                                                                                                                                                         |                           |
| <b>KG-Augmented Drug Target Discovery with LLM Validation</b>   <i>Northeastern University</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <i>Sep 2025 – Present</i> |
| <ul style="list-style-type: none"><li>Combined <b>Personalized PageRank on biomedical knowledge graph</b> with Reciprocal Rank Fusion and LLM-based validation for automated drug target identification across 7 disease configurations; graph-augmented approach improved <b>Precision@10 by 13% over LLM-only baseline</b>.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |
| <b>Domain-Adaptive RAG Pipeline with Hallucination Evaluation</b>   <i>Independent Project</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <i>Sep 2025 – Present</i> |
| <ul style="list-style-type: none"><li>Built a <b>24-configuration factorial evaluation framework</b> over chunking × embeddings × retrievers on BEIR/SciFact (5.2K biomedical abstracts); best config reached <b>nDCG@10 0.723 (+63.9%)</b>. Grounding stage showed retrieval coverage, not ranking precision, is the binding constraint on hallucination. Demonstrates <b>vector database and RAG architecture expertise</b>.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                           |

## BIOMEDICAL NLP & MACHINE LEARNING

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| <b>Signal Propagation on 30M+ Edge Graph (ISMB 2026, 1st Author)</b>   <i>Northeastern University</i> | <i>Sep 2024 – Present</i> |
|-------------------------------------------------------------------------------------------------------|---------------------------|

- Designed algorithms on 30M+ record biomedical knowledge graph with 3 conflict resolution strategies; **systematic ablation across 4,500 configurations**. AUC +27.7% over baseline. Built C++ engine for **8× speedup**. Published at ISMB 2026.

#### LLM Multi-Agent Pipeline for Scientific Literature Extraction | *Northeastern University* *Sep 2025 – Present*

- Three-agent architecture (Extraction → Quality Filter → Judge) processing 372 papers; achieved **94% anchor grounding at \$0.014/paper**; Judge Agent reclassification raised **adjusted F1 from 24% to 83%**. Demonstrates **scalable LLM pipelines for healthcare/biomedical data extraction**.

#### Automated Medical Report Generation (EMNLP 2023, Co-first Author) | *Independent Project* *2023*

- Self-training framework using teacher-student architecture with MedCLIPScore filtering; with 1% labeled data achieved **CIDEr competitive with 100%-data SOTA**; with full data **+20.3% over prior SOTA**. Published at EMNLP 2023.

#### Scaling-Up Medical Vision-Language Learning (EAAI Q1, Co-first Author) | *Jinan University* *2022 – 2023*

- Federated learning framework unifying multi-center medical datasets while preserving patient privacy; **SOTA on all tasks**: image-to-text retrieval R@10 54.9% (+82.4% over baseline), VQA accuracy up to 91.5%.

### SOFTWARE ENGINEERING & CLOUD

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#### Serverless Data Pipeline on AWS | *Independent Project* *2026 – Present*

- S3 → Lambda → SQS → DynamoDB via AWS CDK (Infrastructure-as-Code); **1,000 docs/hour at ~\$0.02/doc**, auto-scales 0–50 concurrent Lambdas, **p99 <100ms**. Demonstrates **cloud-native ML pipeline architecture**.

#### Full-Stack Research Dashboard | *Independent Project* *2026 – Present*

- React + TypeScript + FastAPI + PostgreSQL + Redis + Docker Compose; real-time WebSocket agent status updates, D3.js visualization of decision chain-of-thought, JWT auth, GitHub Actions CI/CD. **500 req/s, 87% test coverage**.

### PUBLICATIONS (8 PEER-REVIEWED)

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**Z. Liu** (1st). ISMB 2026 [AUC +27.7%] | SMARTMiner. EMNLP 2025 [Health Coaching NLP] | **Z. Liu** (1st). SIGIR 2023 | EMNLP 2023 (Co-1st) [Medical Reports] | EAAI Q1 (Co-1st) [Medical VLP] + 3 EI-indexed

### ADDITIONAL

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- **Kaggle Expert (top 0.7%)**: 3 silver medals — NLP, **medical imaging**, finance. Competitive ML applied to healthcare data.
- **Healthcare AI Governance**: Designed auditable LLM outputs with structured citations, hallucination gating, and chain-of-thought reasoning trails for regulatory compliance.
- **Communication**: Presented at ISMB 2026; co-authored across 4 countries (US, UK, China, Singapore). Effective in cross-functional research teams.